Academic Software Requirements Specification (SRS)

Capstone Project: Loan Default Prediction & Credit Risk Scoring — Analyzing the LendingClub Dataset

1. Introduction

1.1 Purpose & Educational Scope

This document outlines the requirements for an academic capstone project designed for Data Analytics students. The objective is to apply data science to FinTech and consumer banking by analyzing historical LendingClub loan data. Students will investigate the primary drivers of loan defaults, evaluate the predictive power of credit scores (FICO) and Debt-to-Income (DTI) ratios, and build a credit risk scoring model.

Students will progress through a complete data analytics lifecycle, transitioning from foundational spreadsheet data validation to advanced programmatic data wrangling, relational database architecture, and interactive business intelligence storytelling.

1.2 Core Learning Objectives & Tech Stack

By completing this project, students will master the following four-pillar tech stack:

- **Excel:** Conduct preliminary data profiling, anomaly detection, and baseline statistical checks on applicant financials and loan terms.
- **Python (Pandas, Scikit-learn, Seaborn):** Automate data cleaning, handle severe class imbalance, engineer credit risk metrics, and build machine learning classification models for default prediction.
- **MySQL:** Design and query a relational database using a Star Schema, reinforcing concepts of financial dimensional modeling, indexing, and SQL aggregations.
- **Power BI:** Model data visually, write DAX measures, and build an interactive executive dashboard that functions as a risk management console for loan officers and bank executives.

2. The Data Pipeline & Tool Application

2.1 Data Sources & Extraction

Students will synthesize borrower demographics, credit history, and loan performance logs:

- **Dataset Sourcing:** Utilize the well-known "LendingClub Loan Dataset" from Kaggle or synthetic equivalent consumer credit datasets.

- **Data Types to Handle:**
    - *Borrower Financials:* Member_ID, Annual_Income, Employment_Length, Home_Ownership, Debt_to_Income_Ratio (DTI), FICO_Score.
    - *Loan Details:* Loan_ID, Loan_Amount, Term (36 or 60 months), Interest_Rate, Installment, Loan_Purpose (e.g., Debt Consolidation, Small Business).
    - *Target & Performance:* Total_Payment, Recoveries, Loan_Status (Fully Paid, Charged Off, Default, Current).

2.2 Phase 1: Initial Data Profiling (Excel)

Before writing code, students must understand the shape and flaws of raw financial data using Excel.

- **Quality Checks:** Use Conditional Formatting to highlight financial impossibilities (e.g., Annual_Income of $\$0$ while holding a large mortgage, or DTI exceeding 100%).
- **Data Validation:** Use Pivot Tables to summarize baseline default rates grouped by Loan_Purpose to identify high-risk borrowing categories (e.g., Small Business vs. Wedding loans).
- **Deliverable:** A "Data Quality Report" summarizing structural flaws, missing employment records, and formatting inconsistencies that need to be addressed programmatically.

2.3 Phase 2: Data Wrangling & Automation (Python)

Students will use Python to handle string parsing, financial data scaling, and target variable creation.

- **String Manipulation:** Strip special characters from continuous variables (e.g., converting interest rates from "15.5%" to a numeric 0.155, or removing "years" from Employment_Length).
- **Target Variable Binarization:** Group the various Loan_Status categories into a clean binary classification target where "Fully Paid" = 0 and "Charged Off" / "Default" = 1. Ongoing/Current loans should be filtered out to avoid data leakage.
- **Handling Missing Values:** Programmatically impute missing Annual_Income or FICO_Score values using the median of the borrower's Home_Ownership or geographic cohort, documenting the statistical rationale in Jupyter Notebook markdown cells.

2.4 Phase 3: Relational Storage (MySQL)

Cleaned datasets must be loaded into a structured database to teach enterprise financial data warehousing concepts.

- **Schema Design:** Build a Star Schema. Create a central Fact_Loans table surrounded by Dim_Borrower, Dim_Loan_Attributes, and Dim_Time (Issue Date).
- **SQL DDL/DML:** Write scripts to CREATE TABLES with strict data types (e.g., DECIMAL(12,2) for monetary values) and insert the cleaned Python outputs.
- **Advanced Querying:** Write JOIN statements linking specific defaults to borrower dimensions. Use Window Functions (AVG(Interest_Rate) OVER (PARTITION BY FICO_Tier ORDER BY Issue_Date)) to track how pricing strategies for different credit tiers evolved over time.

3. Advanced Analytics & Modeling

3.1 Feature Engineering (Python & SQL)

Students will engineer standard consumer credit metrics to assess borrowing capacity mathematically:

- **Income-to-Loan Ratio:** A measure of the borrower's capacity to absorb the requested principal. $$Income\ to\ Loan = \frac{\text{Annual Income}}{\text{Loan Amount}}$$
- **Payment-to-Income (PTI) Ratio:** Evaluating the monthly burden of the new loan. $$PTI = \frac{\text{Monthly Installment}}{\text{Monthly Gross Income}}$$
- **Credit Tier Flag:** Discretizing continuous FICO scores into categorical risk buckets (e.g., Subprime: <620, Fair: $620-679$, Good: $680-739$, Excellent: $740+$).

3.2 Exploratory Data Analysis (Python)

Students will visualize financial risk behaviors and credit distributions using Matplotlib and Seaborn:

- **Distribution Density Plots:** Overlaying the density distributions of FICO_Score for Defaulted loans vs. Fully Paid loans to visually prove that lower credit scores strongly correlate with attrition.
- **Correlation Heatmaps:** Mathematically mapping the relationships between Interest_Rate, Term, Loan_Amount, and the Default_Flag.
- **Categorical Risk Bar Charts:** Comparing the percentage of charge-offs across different Home_Ownership statuses (Rent vs. Mortgage vs. Own).

3.3 Introductory Machine Learning (Python)

Students will apply scikit-learn to solve predictive credit risk problems:

- **Loan Default Prediction (Classification):** Train a Logistic Regression or Random Forest model using FICO score, DTI, loan purpose, and income to predict the binary probability (1 or 0) that an applicant will default.

- **Handling Class Imbalance:** Since defaults typically represent a minority of total loans (e.g., 15%), students must implement techniques like SMOTE (Synthetic Minority Over-sampling Technique) or class weight adjustments to prevent the model from simply predicting "Fully Paid" for every instance.

4. Business Intelligence & Storytelling (Power BI)

4.1 Dashboard Development

The final phase teaches students how to present complex risk metrics to Chief Risk Officers (CROs) and Portfolio Managers.

- **Data Modeling in Power BI:** Connect Power BI to the MySQL database. Re-establish the Star Schema relationships within the Power BI model view.
- **DAX (Data Analysis Expressions):** Write custom measures for Total Default Rate %, Total Capital Exposure ($), Average Interest Rate, and Total Charged-Off Principal.
- **Visual Interface:**
    - **Portfolio Risk KPI Ribbon:** Executive cards displaying total active loans, current portfolio default rate, and average portfolio FICO score.
    - **Risk vs. Return Matrix:** A scatter plot or bubble chart mapping Loan_Purpose categories by their Average Interest Rate (Yield) on the Y-axis against their Default Rate (Risk) on the X-axis.
    - **Geospatial Exposure Map:** A map visual plotting loan originations by State, color-coded by the regional charge-off percentage to identify risky geographic markets.
    - **Interactive Slicers:** Adding filters for users to slice data by FICO Tier, Loan Purpose, Term Length, and Origination Year.

5. Student Submission & Evaluation Rubric

5.1 Repository Structure

Students must submit a logically organized GitHub repository or ZIP file to demonstrate best practices:

- /data: Contains /raw (original files) and /processed (cleaned files ready for MySQL).
- /scripts: Jupyter Notebooks (.ipynb) separated logically:
    - 1_python_data_cleaning_and_string_parsing.ipynb
    - 2_eda_and_credit_feature_engineering.ipynb

    - 3_default_classification_and_smote.ipynb
- /sql: .sql files containing table creation (DDL) and analysis queries.
- /dashboard: The finalized .pbix Power BI file.

5.2 Evaluation Criteria

- **Code Quality & Documentation (30%):** Python code must be commented. SQL queries must use proper indentation and aliases. Jupyter Notebooks must use markdown cells to explain the financial logic behind specific feature engineering (e.g., DTI thresholds).
- **Data Engineering (30%):** Successful execution of the Star Schema in MySQL with enforced Primary/Foreign keys, and accurate conversion of messy text fields into clean numeric formats.
- **Analytical Accuracy (20%):** Correct application of classification for predicting defaults, specifically evaluated using the ROC-AUC score, Precision, and Recall rather than basic Accuracy (to account for the imbalanced dataset).
- **Data Storytelling (20%):** The Power BI dashboard must function as an intuitive risk management console, effectively communicating portfolio exposure and default drivers to banking stakeholders.